

Stakeholder Theory and Agile Software Development

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ABSTRACT

Stakeholder Theory provides models for identifying and mapping stakeholders, and for recognizing who and what really counts in an organization. The objective of this research is to develop a model that helps organizations understand the diversity of stakeholders that influence their product development efforts, provide a means of quantifying the nature of their stake, and understand how agile and lean processes facilitate application of stakeholder theory in software product development.

Categories and Subject Descriptors

D.2.9 [Software Engineering]: Management – *Software process models*.

K.6.3 [Management of Computing and Information Systems]: Software Management – *software development, software process*.

General Terms

Management, Human Factors, Theory.

Keywords

stakeholder, agile, lean, stakeholder mapping, stakeholder groups

1. RESEARCH AREA

There is a body of knowledge that exists in Agile and Lean practices, organization patterns, and systems design that can help teams become more productive and understand better the context in which they are creating products. Separately, there is a body of knowledge in stakeholder theory that helps organizations understand the needs and influences of different stakeholder groups. A particular focus of my research is on the application of stakeholder concepts to software development, and specifically to geographically distributed product development organizations.

2. PROBLEM STATEMENT

How can managers of an organization better identify and engage with the stakeholders that influence, and are influenced by, its product development efforts? How do agile and lean principles help the application of stakeholder concepts to a product development effort or even to product architecture?

In other words, can we develop a model based on stakeholder concepts that helps us to identify and engage with the stakeholders that influence, and are influenced by a product development effort or a system's architecture?

3. LITERATURE REVIEW

Stakeholder Theory is an area of strategic management that provides models for identifying and mapping stakeholders [1, 2]. Freeman identifies seven techniques or strategies for creating value for stakeholders, including stakeholder assessment, stakeholder behavior analysis, understanding stakeholders in more depth, assessing stakeholder strategies, developing specific strategies for stakeholders, creating new models of interaction for stakeholders, developing integrative value creation strategies [2]. The diagram in Figure 1 shows a typical stakeholder map using Freeman's two-tier stakeholder model. This map shows the firm at the center, and shows the primary and secondary stakeholder groups in the firm.

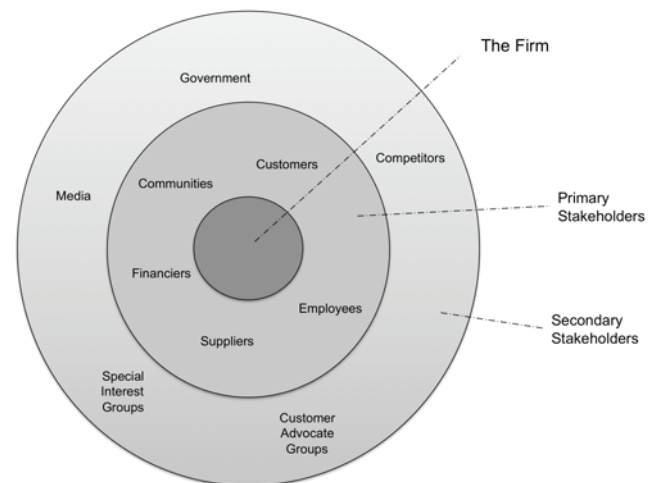
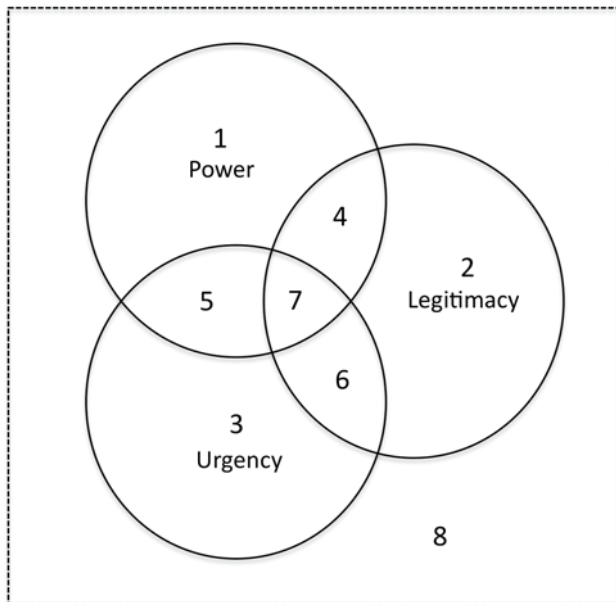


Figure 1 Two-tier stakeholder map [2]

Some stakeholders will naturally have a higher degree of relative importance or urgency. Saliency, the principle of defining who and what really counts to an organization, is key to understanding the stakeholders that managers should pay most attention to. Mitchell, Agle and Wood define a model for classifying stakeholders based on attributes of power, legitimacy and urgency that helps managers understand and quantify the degree of salience a stakeholder possesses [3]. The category of salience of a

stakeholder or stakeholder group can change over time. Therefore, salience has a dynamic aspect. The diagram in Figure 2 shows the different qualitative classes of stakeholders as described by Mitchell's model.



Latent stakeholders possess one attribute

- (1) **Dormant** stakeholders have Power
- (2) **Discretionary** stakeholders have Legitimacy
- (3) **Demanding** stakeholders have Urgency

Expectant stakeholders possess two attributes

- (4) **Dominant** stakeholders have Power and Legitimacy
- (5) **Dangerous** stakeholders have Power and Urgency
- (6) **Dependant** stakeholders have Legitimacy and Urgency
- (7) **Definitive** stakeholders possess all three attributes
- (8) **Non-stakeholders** possess none of the attributes

Figure 2: Qualitative classes of stakeholders

Phillips brings a deeper ethical dimension to the principles of stakeholder identification, discussing concepts of fairness when dealing with stakeholders [4].

Although the term stakeholder is often used in software development, it is predominantly limited to customers, end-users, or project sponsors. They do not focus on product development teams as a diverse group of stakeholders with diverse interests. Where there is deeper mention of stakeholders, they typically treat the development team or 'the customer' as a single stakeholder. As far as I can tell, there is currently no formal use of Stakeholder Theory applied to software product development teams, and software development processes in general. I have so far been unable to identify any application of stakeholder theory in the context of agile and lean development. A specific focus of my research is the application of stakeholder theory to agile and lean processes.

4. CURRENT RESULTS

Pilot interviews and subsequent data analysis is helping to refine the research instruments, and overall research objective. From the research data I have begun to map stakeholders in a product development organization to different stakeholder groups, and to specific salience categories. I have begun to understand the circumstances that cause stakeholders to transition from one category to another, and identify recommendations. This is helping further refine the overall research questions. In addition, this research considers non-human stakeholders such as the architecture of a product or system, and examines the stakeholders that influence, or are influenced by, the architecture [5].

5. EXPECTED CONTRIBUTIONS OF THIS RESEARCH

This research will develop a model that helps managers understand the broad range of stakeholders that influence, and are influenced by, its activities. This model can be applied to the firm itself, or to units within the firm, such as an individual business unit or project team. More fundamentally, this model can be applied to a software product development effort to help managers understand who and what really counts when developing products. The model includes strategies for identifying and engaging with stakeholders, and for classifying stakeholders by mapping them to one of five stakeholder groups. The research will further show, in the specific context of how agile and lean processes enable and facilitate stakeholder engagement. The model further includes a means of classifying the salience of each stakeholder or stakeholder group, and for understanding how and when stakeholders can transition from one salience category to another. This research will relate how agile processes provide a means to control this salience transition and to prevent or affect it.

6. RESEARCH SUPERVISOR

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7. REFERENCES

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